FREE RESOURCE | THE MBA PA

Conduit Quality Assurance System

A Framework for Building Conduit Quality Programs That Survive Turnover

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The MBA PA

A comprehensive system for cardiac surgery programs to define, measure, and maintain conduit quality — including grading standards, a four-stage harvester competency ladder, quality dashboards, a turnover survival protocol, and a program self-assessment across seven dimensions.

How to Use This System

- **For program leaders:** Use the grading standards and competency ladder to establish objective, written criteria for conduit quality. Eliminate subjective "good enough" assessments.
- **For harvesters and preceptors:** Use the ladder to track individual progression and identify specific skill gaps before they affect patient outcomes.
- **For quality committees:** Use the dashboard metrics to monitor program-level conduit quality trends and detect deterioration before it becomes a pattern.
- **When harvesters leave:** Activate the turnover survival protocol to maintain quality standards during transitions — the system, not any individual, protects outcomes.

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1) Conduit Quality Grading Standards

Define what "good" looks like — objectively, consistently, across all harvesters.

Without written grading criteria, conduit quality assessment is subjective — "it looked fine" is not a quality standard. Use the following framework to establish program-wide grading for endoscopic vein, radial artery, and open conduit harvesting.

CONDUIT	GRADE A — EXCELLENT	GRADE B — ACCEPTABLE	GRADE C — NEEDS IMPROVEMENT
EVH (Vein)	No visible thermal injury. Intact adventitia. No side-branch avulsions. Conduit length adequate for all targets. Harvest time within program benchmarks.	Minor adventitial disruptions ≤2. One small thermal mark. Side branches clipped cleanly. Length adequate. Time within 120% of benchmark.	Multiple thermal marks, adventitial tears, or branch avulsions. Segment loss requiring additional harvest. Time exceeds 150% of benchmark.
Radial Artery	No palpable spasm at harvest completion. Intact endothelium. Adequate length with clean pedicle. Papaverine response documented.	Mild spasm resolving with papaverine. Minor adventitial disruption. Pedicle largely intact. Length adequate.	Persistent spasm. Endothelial injury visible. Uncontrolled branching. Segment loss or conversion required.
Open Vein	Clean dissection with minimal adventitial handling. No scalpel or scissor injuries to conduit. Side branches ligated cleanly. Wound closure optimal.	Minor handling marks. One small adventitial nick without full-thickness injury. Branches controlled. Acceptable wound.	Multiple handling injuries. Full-thickness defects. Uncontrolled branches. Excessive wound tension or poor closure.

Implementation: Assign a grade to every harvested conduit. Track grades by harvester. Review grade distributions monthly. A harvester whose Grade C rate exceeds 10% needs targeted remediation — not more autonomy.

Harvester Competency Ladder & Quality Dashboard

2) Four-Stage Harvester Competency Ladder

Written gate criteria for advancing from observer to independent harvester — with preceptor validation at every stage.

1

Observer — "See One"

Observes ≥20 harvests across all conduit types (EVH vein, radial, open). Demonstrates knowledge of equipment setup, troubleshooting common device errors, and anatomy review. Passes written anatomy and device competency quiz. **Gate criteria:** Preceptor signs off on equipment competency and anatomy knowledge. Minimum 20 observed cases documented.

2

Supervised Harvester — "Do One With Help"

Performs ≥30 harvests under direct preceptor supervision. Preceptor is scrubbed and immediately available. Focus on developing tactile feel, spatial awareness, and branch management. **Gate criteria:** ≥90% Grade A or B conduits over last 10 cases. Preceptor confirms readiness for independent progression. Average harvest time within 130% of program benchmark.

3

Provisional Independent — "Teach One"

Independent harvesting with preceptor available in-house for backup. Performs ≥50 cases independently. Begins precepting Stage 1-2 harvesters under supervision. **Gate criteria:** ≥95% Grade A or B conduits. Zero unplanned conversions to open in last 20 cases. Preceptor evaluation of precepting readiness. Lead APP review of grade distribution.

4

Master Harvester / Preceptor — "See One, Do One, Teach One"

Full independent practice. Qualified to precept all stages. Serves as first-call backup for difficult harvests. Participates in monthly quality review. Maintains ≥95% Grade A or B conduits. **Ongoing requirement:** Quarterly grade audit — if Grade C rate exceeds 5%, automatic review with Lead APP. Annual preceptor re-certification.

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3) Conduit Quality Dashboard

Monthly metrics and thresholds for program-level quality monitoring.

Track these metrics monthly. Review at program quality meetings. Trends matter more than any single month's data. Set program-specific thresholds and escalate when thresholds are breached for two consecutive months.

Monthly Dashboard Template

METRIC	TARGET	THRESHOLD	CURRENT MONTH
Overall Grade A rate	≥ 70%	< 60%	___%
Grade C rate	≤ 5%	> 10%	___%
Unplanned conversion rate	≤ 2%	> 5%	___%
Avg harvest time (EVH)	≤ 25 min	> 35 min	___ min
Conduit-related OR delays	0/month	≥ 2/month	___
Harvester grade distribution reviewed	Monthly	Quarterly	Yes / No
Preceptor feedback completed	Per case	Monthly	___ / ___
Stage 1-2 harvester progression on track	100%	< 80%	___%

Turnover Survival Protocol & Program Self-Assessment

4) Turnover Survival Protocol

What to do when a harvester leaves — before quality deteriorates.

Conduit quality is the most vulnerable part of a cardiac surgery program when a harvester leaves. Without a protocol, programs enter reactive mode: remaining harvesters absorb the volume, fatigue increases, and quality erodes silently until a conduit fails. This protocol is designed to be activated the moment a resignation is received — not after the departure.

Turnover Survival Checklist

- [ ] **Day 0 — Resignation Received:**
  Notify Lead APP and CVOR Director. Flag departing harvester's caseload for redistribution review. Pull last 3 months of grade data — identify any quality trends that need immediate attention. Schedule knowledge-transfer sessions for any program-specific techniques or preferences.

- [ ] **Week 1 — Transition Planning:**
  Map all harvests over next 90 days to remaining harvesters. If capacity gap exceeds 20%, activate locum or cross-coverage plan immediately. Do not wait until the person leaves. Notify perfusion and anesthesia of any anticipated schedule adjustments. Schedule departing harvester to co-harvest with their replacement or next-in-line for at least 3 cases.

- [ ] **Month 1 Post-Departure — Quality Surveillance:**
  Increase conduit grading frequency to every case (if not already). Run weekly grade distribution reports — do not wait for month-end. Any Grade C triggers same-day preceptor review. Monitor harvester fatigue: if any remaining harvester exceeds 120% of baseline volume, escalate to CVOR Director.

- [ ] **Months 2–3 — Stabilization:**
  Return to monthly grade audits when quality thresholds are met for 4 consecutive weeks. If new hire onboarding is underway, Stage 1-2 harvesters should not be used to fill volume gaps — this risks both conduit quality and trainee development. Complete post-turnover quality review with CVOR Director, Lead APP, and remaining harvesters.

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5) Program Self-Assessment

Score your program across seven dimensions of conduit quality readiness.

Rate each dimension 1–5. Be honest — the goal is to identify the gap between where you are and where you need to be.

Seven Dimensions of Conduit Quality Readiness

1 **Written Grading Standards**
Do you have objective, written criteria for grading every harvested conduit — or does assessment depend on who's looking? Score: ___ / 5

2 **Harvester Competency Documentation**
Is each harvester's competency level formally tracked — with gate criteria, preceptor sign-offs, and progression milestones? Score: ___ / 5

3 **Monthly Quality Metrics**
Are conduit quality metrics reviewed at least monthly at a program quality meeting — or does quality review happen only when something goes wrong? Score: ___ / 5

4 **Preceptor Accountability**
Are preceptors formally assigned, trained in feedback delivery, and evaluated on precepting outcomes — or is precepting an unfunded mandate? Score: ___ / 5

5 **Remediation Pathway**
When a harvester's quality deteriorates, is there a structured remediation process — or does the program just hope it improves? Score: ___ / 5

6 **Turnover Resilience**
Is there a documented protocol for maintaining quality during harvester transitions — or does the program enter crisis mode with every departure? Score: ___ / 5

7 **Quality Culture**
Does your program culture treat conduit quality as a shared team responsibility — or as an individual harvester problem? Do harvesters feel safe reporting quality concerns? Score: ___ / 5

Composite Score: ___ / 35

28–35	Quality-Systematized. Your program has structural conduit quality assurance.
21–27	Quality-Aware. Good practices exist but they're person-dependent. System gaps remain.
14–20	Quality-Vulnerable. Conduit quality depends on individual harvesters without system backup.
Below 14	Quality-at-Risk. Program has no structured conduit quality assurance. Every harvester departure is a patient safety event waiting to happen.

Next Steps: From Assessment to Action

1. **Start with the lowest-scoring dimension.** A single systematized dimension reduces more risk than partial improvement across all seven.

2. **Write it down.** If your grading standards exist only in the head of your most experienced harvester, they don't exist. Document first, implement second.

3. **Make quality review routine — not reactive.** Monthly dashboard reviews transform conduit quality from an incident-driven conversation into a managed process.

4. **Build before you need it.** The turnover survival protocol must be in place before a resignation arrives. By the time someone gives notice, it's too late to design the response.

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This is an operational systems framework, not clinical teaching or patient-care directives. It draws on established quality-improvement methodology applied to the cardiac surgery domain.